

NATHAN WONG SHIH HAO 黄诗豪

+852 9745 0024 | nathanwongshihhao@gmail.com | linkedin.com/in/nwsh | nathanwongsh.dev

EDUCATION

The Hong Kong University of Science and Technology, Hong Kong September 2022-June 2026
Bachelor of Engineering in Computer Science (Exchange to University of Waterloo)
• **Awards:** Reaching Out Award (2024), University Entrance Scholarship (2022)

WORK EXPERIENCE

Capgemini Hong Kong | Technology Consultant | Hong Kong March 2026-Present
• Conducted independent research and benchmarking of Small Language Models (SLMs), evaluating on-device inference performance across edge hardware targets including latency, hallucinations, memory footprint, and accuracy trade-offs.
• Designed an evaluation framework to benchmark 5+ SLMs (e.g. Phi, Gemma, Qwen) against full LLMs across domain-specific tasks, using a live dashboard to present findings that advised the team's edge AI deployment strategy.
• Built and deployed a proof-of-concept SLM/LLM router using Python and FastAPI, implementing four dynamic routing strategies including a fine-tuned sentence transformer based on query complexity classification to optimize cost by 30%.

Hitachi Rail GTS | R&D Software Intern | Hong Kong June 2025-February 2026
• Enhanced the performance of the primary scripting algorithm by 20% and optimized the data collection system through the use of batch processing in ClickHouse DB, leveraging its OLAP architecture for improved query performance.
• Coordinated the deployment of Project NAIA to Hitachi's ALVEA platform by implementing a Jenkins CI/CD pipeline for integrated testing and employing a Kubernetes cluster for operational management, with an emphasis on scalability.
• Engineered a Kafka pipeline and implemented a producer-consumer side schema registry, integrating schema validation to facilitate the ingestion of data in various formats for improved data processing and pipeline integration.
• Developed ARIMA, SARIMAX, and LightGBM models for Project NAIA, achieving an average 97% R² on forecasting.

Total Rehabilitation Management (TRM HK) | Software Developer | Hong Kong June 2024-August 2024
• Designed and implemented a client-side search bar using JavaScript, C#, and MySQL, reducing search time by 50%.
• Evaluated multiple large language models (GPT-4o, GPT 4o-mini, Claude, etc.) to filter and prioritize insurance claims.
• Developed an internal staff chatbot, increasing engagement from 2 messages per week to 20 messages per day.

LEADERSHIP & ACTIVITIES

HKUSTSU Southeast Asian Students' Association | President | Hong Kong February 2023-February 2024
• Provided recommendations to Southeast Asian students in HKUST and acting as representatives to the Student Union.
• Increased participation rates to events by 300% from the 4 major Southeast Asian country groups in HKUST.

ICMS | 13th Internal Affairs Executive Director | Hong Kong February 2023-February 2024
• Led the Internal Affairs division consisting of 28 chapter directors and 92 associates from all over the globe.
• Established relations with companies such as Axiata, Shopee, Shell, Procter & Gamble, Maybank and Sunway.

PROJECTS

Morning Digest Using Github Actions
• Engineered an AI-powered pipeline using Github Actions, Google OAuth, Gmail, GDrive for personalized daily briefs.
• Delivered a scheduled daily reporting system that cut information overload and improved early-day productivity by 30%.

Containerized Local Small Language Model Inference Using Ollama
• Deployed quantized SLMs in GGUF formats via Ollama to optimize inference efficiency on consumer-grade hardware.
• Containerized Ollama using Docker to enable reproducible, environment-agnostic local SLM inference.

Time Series Forecasting
• Initiated a self-directed spinoff project focused on gaining insights into time series forecasting, centered at S&P 500.
• Evaluated various time series forecasting methods, including ARIMA, SARIMA, ETS, LightGBM and LSTMs.

Optical Character Reader (OCR) and Summarizer
• Extracted and summarized data from PDF documents, specifically oriented for interpreting doctor handwriting.
• Specialized in prompt engineering to create and test optimal prompts for effective information retrieval.

SKILLS AND COMPETENCIES

• **Coding Languages:** Proficient in Python, C, C++, C#, Java, Scala, JavaScript, MySQL, HTML, CSS, Bash.
• **Tools:** Git, Docker, Jenkins, Kubernetes, LM Studio, Ollama, Apache Spark, Hadoop, Kafka, Avro and Pulsar.
• **Other Skills:** Cloud Computing (AWS, Azure, GCP), DevOps, Data Science, Time Series Forecasting, Networking.
• **Languages:** Fluent in English, Bahasa Indonesia, Chinese (Mandarin).
• **Interests:** Golfing, travelling, landscape photography, Southeast Asian economies.